Eigenvalues & Eigenvectors

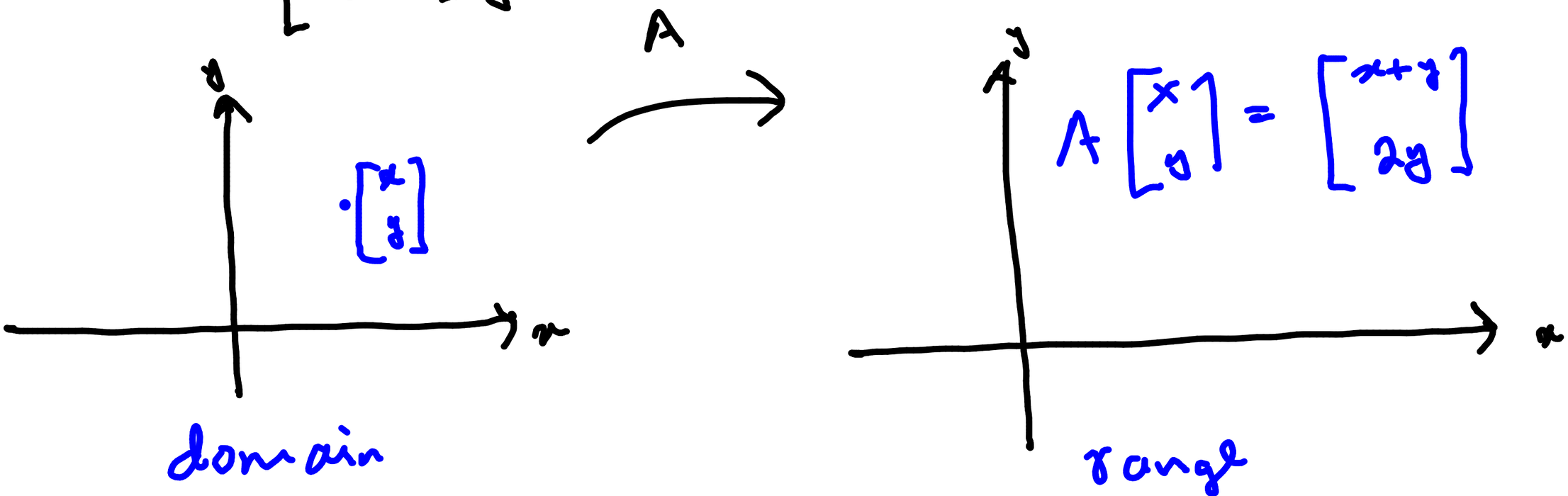
- Matrix as a map from one space to another] Linear Transformation
- Eigenvalues and Eigenvectors
- Eigenbases, Diagonalization, Quadratic forms

Reference: ***Chapter 8, Advanced Engineering Mathematics, Kreyszig***

Matrix is a map from one space to another.

Or in more simple terms, matrix multiplication transforms a vector.

e.g. $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$



$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ 2y \end{bmatrix}$$

Q. $A = \begin{bmatrix} 1 & 5 & 6 \\ 0 & 2 & 5 \end{bmatrix}$ associate A as a map from one space to another

$$\begin{bmatrix} 1 & 5 & 6 \\ 0 & 2 & 5 \end{bmatrix} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\text{3d input}} = \underbrace{\begin{bmatrix} x + 5y + 6z \\ 2y + 5z \end{bmatrix}}_{\text{2d output}}$$

Q.

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 0 & 1 \\ -1 & -1 & 2 \\ -5 & 1 & \pi \end{bmatrix}$$

$$A : (\,???\,) \longrightarrow (\,???\,)$$

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 0 & 1 \\ -1 & -1 & 2 \\ -5 & 1 & \pi \end{bmatrix} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\text{3 dimensional input}} = \underbrace{\begin{bmatrix} x + 2y + 5z \\ z \\ -x - y + 2z \\ -5x + y + \pi z \end{bmatrix}}_{\text{4 dim output}}$$

A matrix M represents a function

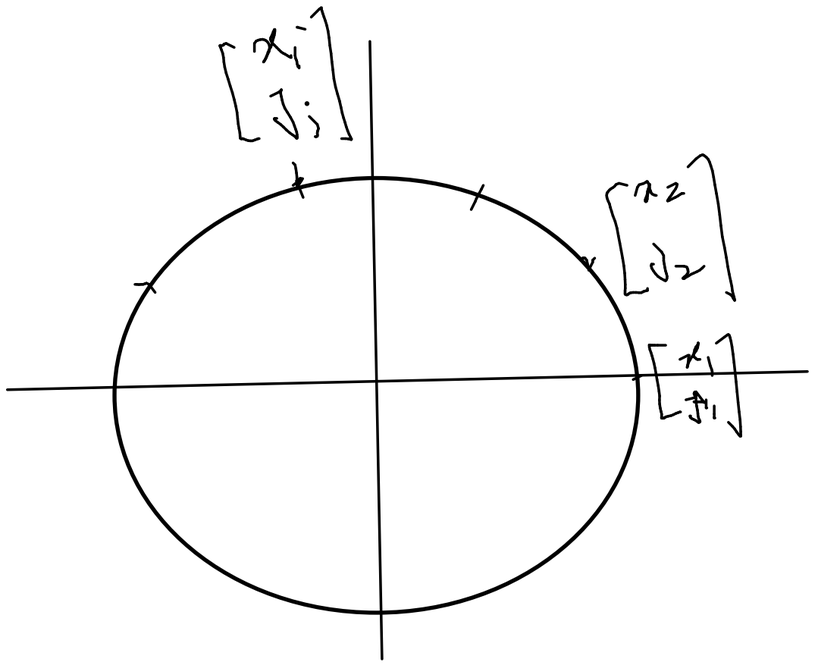
with $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ as input

& $M \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ as output

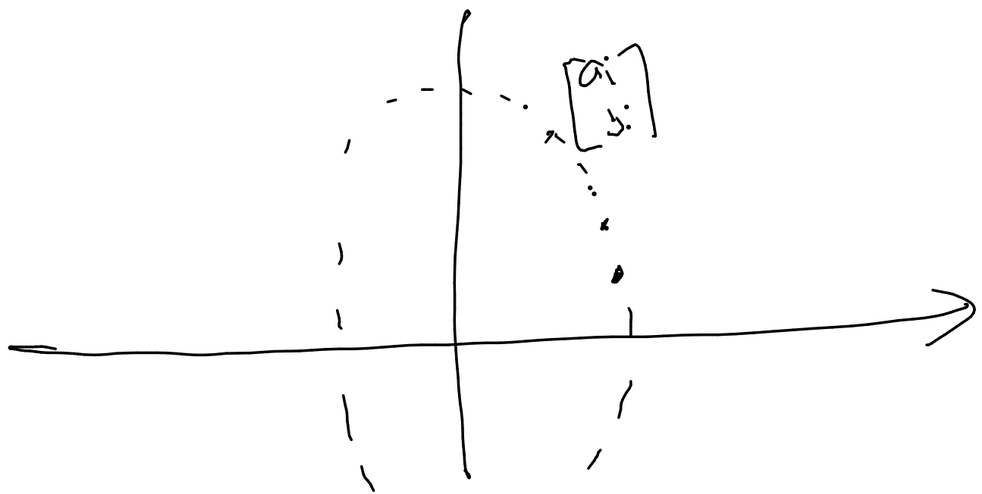
Matlab exercise:

Take 100 uniformly spaced points on the unit circle centered at origin. Then take transformation of the each of those points by left multiplication by the matrix below. Join these new points to find the transformation of the unit circle by this matrix.

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$$



$\rightarrow$ find $\begin{bmatrix} a_i \\ b_i \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix}$

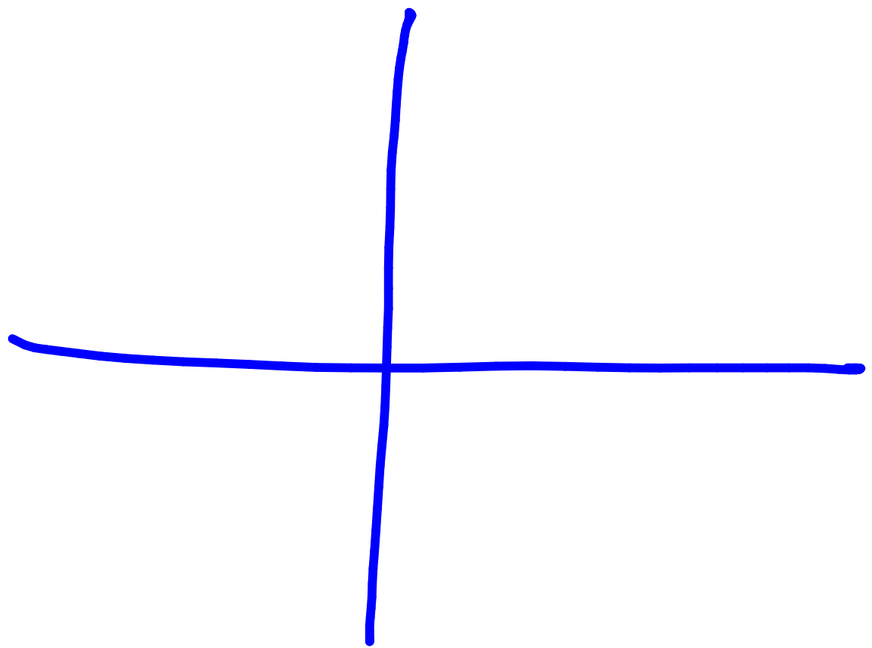


$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}, \begin{bmatrix} x_3 \\ y_3 \end{bmatrix} \quad \cdots\cdots \quad \begin{bmatrix} x_{100} \\ y_{100} \end{bmatrix}$$

$$\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = A\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \quad \begin{bmatrix} a_2 \\ b_2 \end{bmatrix} = A\begin{bmatrix} x_2 \\ y_2 \end{bmatrix}, \quad \cdots, \quad \begin{bmatrix} a_{100} \\ b_{100} \end{bmatrix} = A\begin{bmatrix} x_{100} \\ y_{100} \end{bmatrix}$$

OK??

$$\underset{2\times 2}{A}\ \underset{2\times 100}{\begin{bmatrix} x_1 & x_2 & x_3 & \cdots & x_{100} \\ y_1 & y_2 & y_3 & \cdots & y_{100} \end{bmatrix}} = \begin{bmatrix} A\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} & A\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} & \cdots & A\begin{bmatrix} x_{100} \\ y_{100} \end{bmatrix} \end{bmatrix}$$

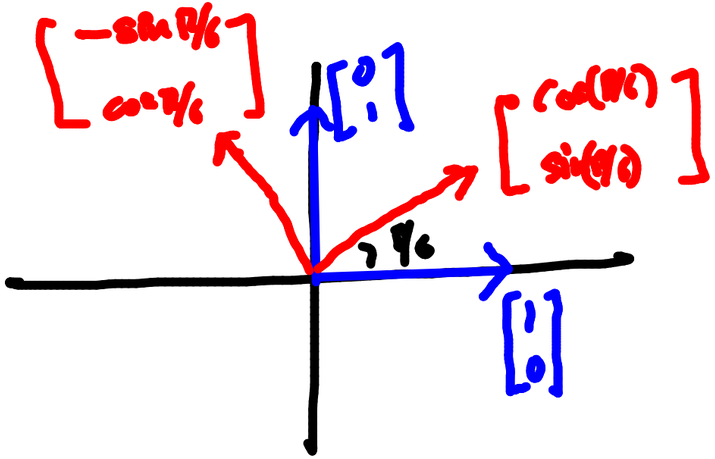


$$A = \begin{bmatrix} * & * \\ * & * \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}, \quad A = \begin{bmatrix} 100 & 0 \\ 0 & 0.1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

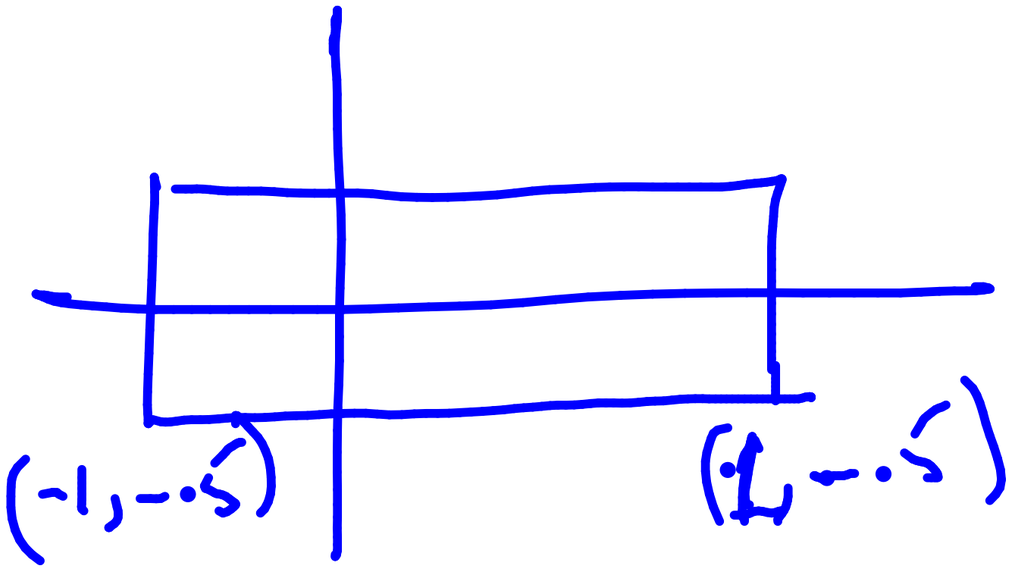
Q. find a 2x2 matrix which rotates $\begin{bmatrix}1\\0\end{bmatrix}$ & $\begin{bmatrix}0\\1\end{bmatrix}$ by $\pi/6$ anticlockwise ?



$$\begin{bmatrix}a\\c\end{bmatrix} = \begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}\cos\pi/6\\ \sin\pi/6\end{bmatrix}$$

$$\begin{bmatrix}b\\d\end{bmatrix} = \begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}-\sin\pi/6\\ \cos\pi/6\end{bmatrix}$$

$$A = \begin{bmatrix}\cos\pi/6 & -\sin\pi/6\\ \sin\pi/6 & \cos\pi/6\end{bmatrix}$$


(-1, -.5)
(1, -.5)

Q:

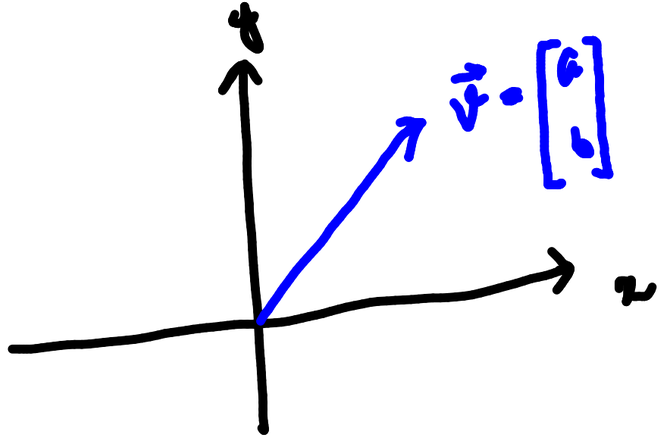
$$A = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} \cos \pi/6 & -\sin \pi/6 \\ \sin \pi/6 & \cos \pi/6 \end{bmatrix}$$

$$AB \begin{bmatrix} x \\ y \end{bmatrix} = ??$$

stretch & rotate

Last class.

Q.



$A\begin{bmatrix} a \\ b \end{bmatrix} =$ rotation of any $\vec{v}$ by an angle θ anticlockwise

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Q.

$$A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$ 1st column of A

$$A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$ 2nd column of A

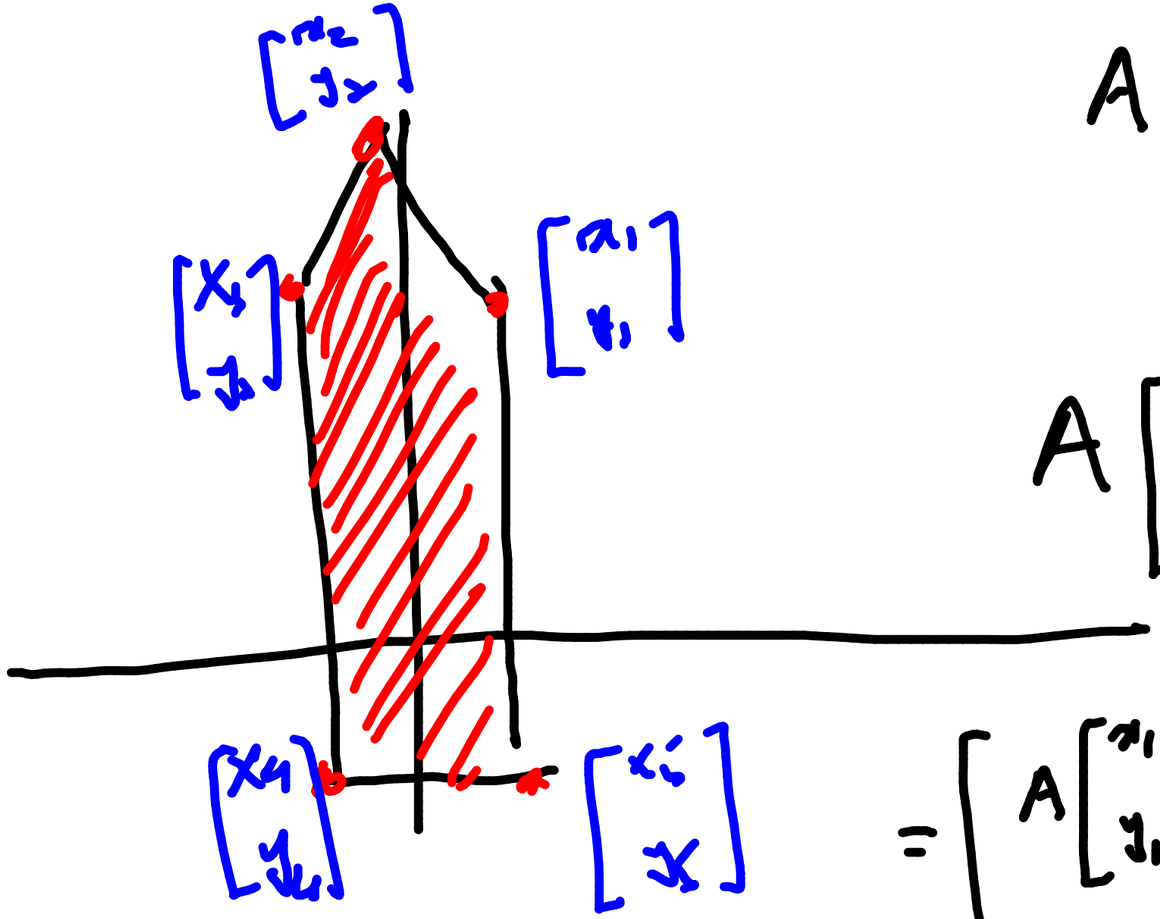
$\vec{v} = \begin{bmatrix} a \\ b \end{bmatrix}$, find rotation of $\begin{bmatrix} a \\ b \end{bmatrix}$ by θ.

, say $\vec{u} = \begin{bmatrix} c \\ d \end{bmatrix}$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} c \\ d \end{bmatrix}$$

$$A = \begin{bmatrix} \cos(\pi/6) & -\sin(\pi/6) \\ \sin(\pi/6) & \cos(\pi/6) \end{bmatrix} \qquad \vec{v} = \begin{bmatrix} a \\ b \end{bmatrix} = a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad \theta = \pi/6$$

$$A\begin{bmatrix} a \\ b \end{bmatrix} = A\left(a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \underbrace{a\left(A\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + b\left(A\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)}$$



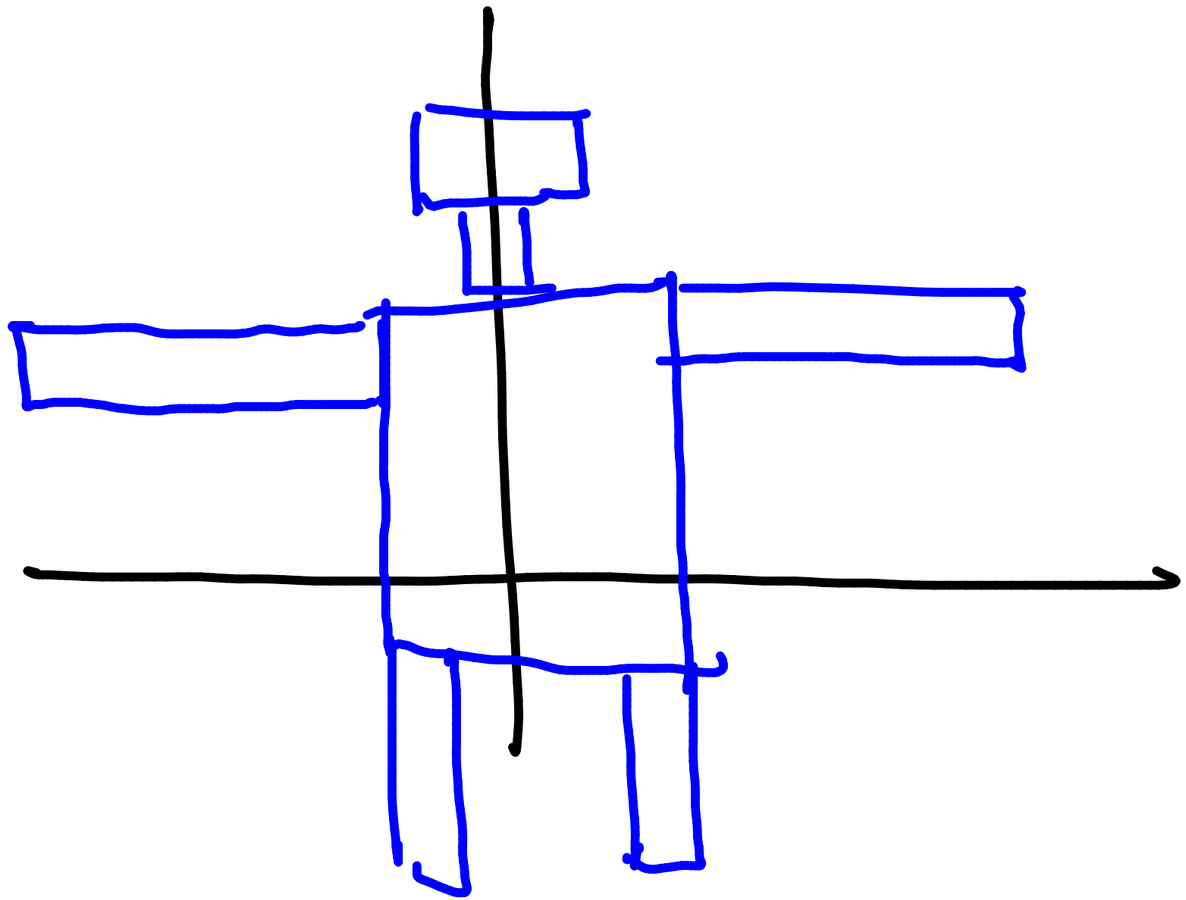
$$A = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

$$A \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ y_1 & y_2 & y_3 & y_4 & y_5 \end{bmatrix}$$

$$= \begin{bmatrix} A\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} & A\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} & \cdots & A\begin{bmatrix} x_5 \\ y_5 \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} \begin{bmatrix} \text{rotation} \\ \text{of} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \end{bmatrix} & \begin{bmatrix} \text{rotation} \\ \text{of} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \end{bmatrix} & \cdots & \end{bmatrix}$$

Q. draw a shape like



& animate it to rotate